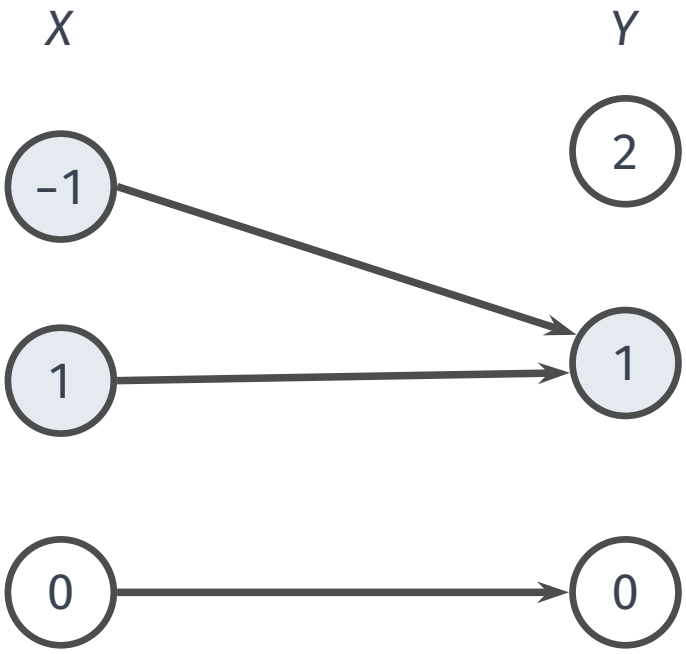
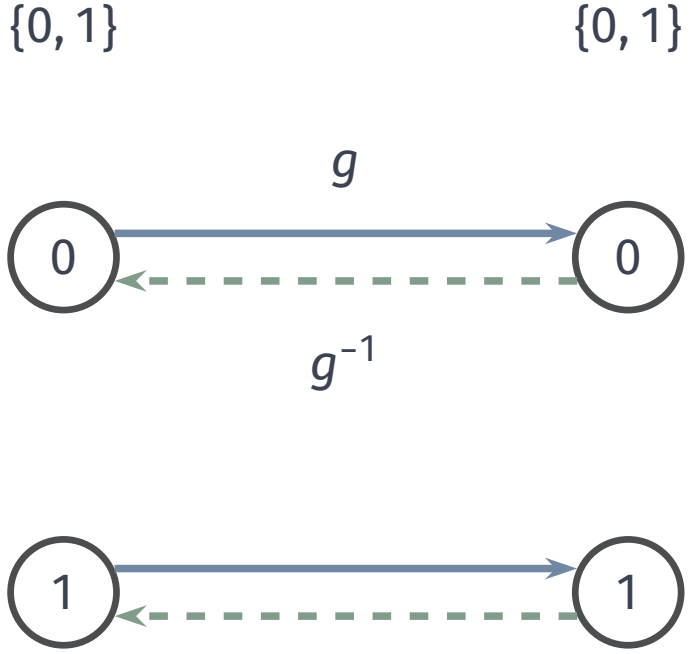
$f(x) = x^2$



$f^{-1}(\{1\}) = \{-1, 1\}$

$g(x) = x^2$



$g^{-1}(1) = 1$